KOM3792 CONTROL LABORATORY

DC MOTOR POSITION CONTROL

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EXPERIMENT OBJECTIVES

The objective in this experiment is to introduce the student to the fundamentals of control using the PID family of compensators. At the end of this session, you should know the following:

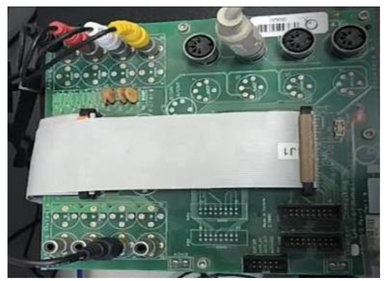
- How to mathematically model the servo plant from first principles.
- An understanding of the different tuning parameters in the controller.
- To design and simulate a PV controller to meet the required specifications.
- To implement your controller and evaluate its performance.

SYSTEM REQUIREMENTS

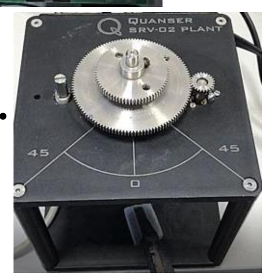
To complete this lab, the following hardware is required:

- Quanser UPM 2405/1503 Power Module or equivalent.

- Quanser Q4 DAQ.



- Quanser SRV02 servo plant.



- PC equipped with the required software

Wiring and Connections: The first task upon entering the lab is to ensure that the complete system is wired as described in the Experiment #0 - Introduction. It is assumed that the student has the knowledge on Experiment #0.

MATHEMATICAL MODEL OF DC MOTOR

A common actuator in control systems is the DC motor. It directly provides rotary motion and, coupled with wheels or drums and cables, can provide translational motion.

We shall begin by examining the electrical component of the motor first. In Fig.1, you see the electrical schematic of the armature circuit.

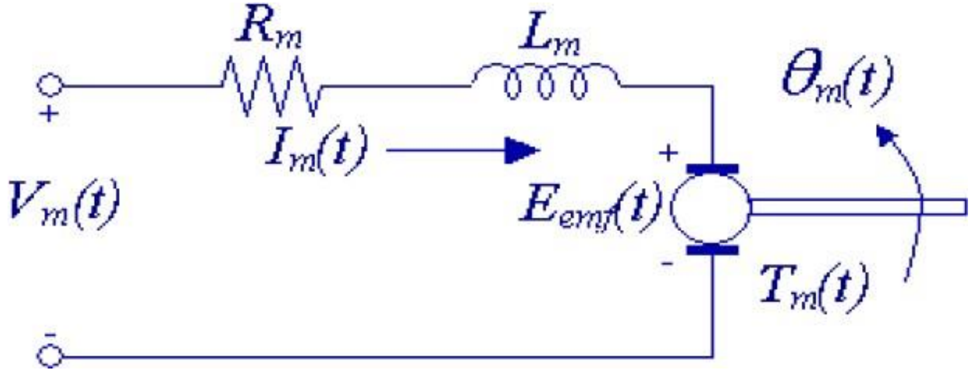


Fig.1 Armature circuit in the time-domain

MATHEMATICAL MODEL OF DC MOTOR

Using Kirchhoff's voltage law, we obtain the following equation:

$$V_m - R_m I_m - L_m \frac{dI_m}{dt} - E_{emf} = 0$$

Since $L_m \ll R_m$ we can disregard the motor inductance leaving us with:

$$I_m = \frac{V_m - E_{emf}}{R_m}$$

We know that the *back-emf* created by the motor is proportional to the motor shaft velocity ω_m such that

$$I_m = \frac{V_m - K_m \dot{\theta}_m}{R_m} \qquad (\dot{\theta}_m = \omega_m) \tag{1}$$

MATHEMATICAL MODEL OF DC MOTOR

We now shift over to the mechanical aspect of the motor and begin by applying Newton's 2nd law of motion to the motor shaft:

$$J_m \ddot{\theta}_m = T_m - \frac{T_l}{\eta_g K_g} \tag{2}$$

where $\frac{T_l}{\eta_g K_g}$ is the load torque seen thru the gears. And η_g is the efficiency of the gearbox.

We now apply the 2nd law of motion at the load of the motor:

$$J_l \ddot{\theta}_l = T_l - B_{eq} \dot{\theta}_l \tag{3}$$

where B_{eq} is the viscous damping coefficient as seen at the output. Substituting (2) into (3), we are left with:

$$J_l \ddot{\theta}_l = \eta_g K_g T_m - \eta_g K_g J_m \ddot{\theta}_m - B_{eq} \dot{\theta}_l \tag{4}$$

MATHEMATICAL MODEL OF DC MOTOR

We know that $\theta_m = K_g\theta_l$ and $T_m = \eta_m K_t I_m$, we can rewrite (4) as:

$$J_l\ddot{\theta}_l + \eta_g K_g^2 J_m \ddot{\theta}_l + B_{eq}\dot{\theta}_l = \eta_g \eta_m K_g K_t I_m \quad (4)$$

Finally, we can combine the electrical and mechanical equations by substituting (1) into (4), yielding our desired transfer function:

$$\boxed{\frac{\theta_l(s)}{V_m(s)} = \frac{\eta_g \eta_m K_t K_g}{J_{eq} R_m s^2 + (B_{eq} R_m + \eta_g \eta_m K_m K_t K_g^2)s}}$$

where $J_{eq} = J_l + \eta_g J_m K_g^2$. This can be interpreted as the being the equivalent moment of inertia of the motor system as seen at the output.

MATHEMATICAL MODEL OF DC MOTOR

SRV02 Nomenclature

Symbol	Description	Nominal value	Unit
Rm	Armature resistance	2.6	Ω
Km	Back-emf constant	0.007678	Vs/rad
Kt	Motor-torque constant	0.007683	Nm/A
Jm	Motor moment of inertia	3.87e-7	$kg.m^2$
Jeq	Equivalentmoment ofinertia at theload	2.084e-3	$kg.m^2$
Beq	Equivalent viscous damping coefficient	4.0e-3	
Kg	SRV02 system gear ratio (motor->load)	70 (14x5)	
η_g	Gearbox efficiency	0.9	
η_m	Motor Efficiency	0.69	

CONTROLLER

The structure of the control system has the form shown in the Fig.3 below.

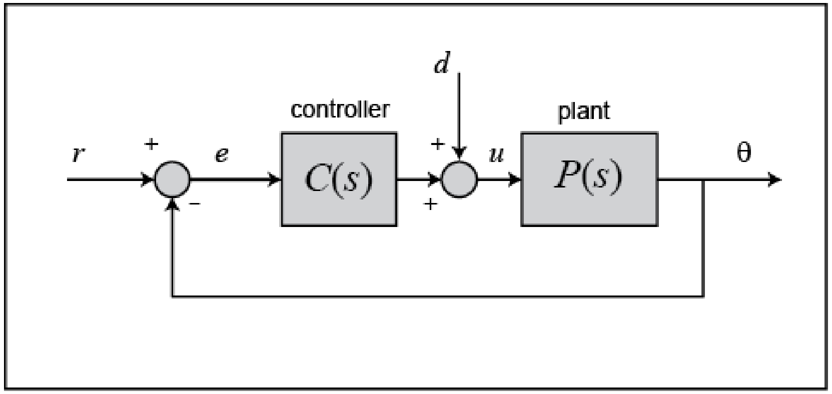


Fig. 3 The structure of the control system

Desired specifications are:

- The overshoot should be less than 5% ($\zeta > 0.707$).
- The time to first peak should be 0.1s $\left(T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}\right)$

CONTROLLER

This lab involves designing a PV (Position-Velocity) controller for the servo plant. The PV controller that will be implemented in this lab has the form:

$$\boxed{V_m = -K_p(\theta - \theta_d) - K_v\dot{\theta}}$$

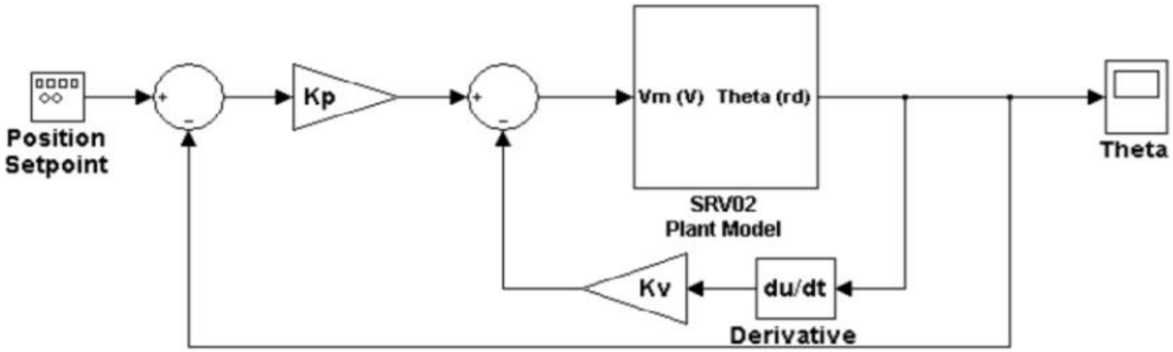


Fig.4 PV Controller for the SRV02 Plant

CONTROLLER

The purpose of this lab is to design a controller using the 2nd order transfer function representation of the form given:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$ *closed loop transfer function*

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$ *characteristic equation*

! ***Show the steps for finding the coefficients of controller (K_p, K_v) to meet desired specifications in the experiment report.***

★ ***Hint: You can use root locus approach or pole placement***

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DC MOTOR SPEED CONTROL

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EXPERIMENT OBJECTIVES

The objective in this experiment is to design a controller that will regulate the speed of the output shaft.

At the end of this session, you should know the following:

- An understanding of the different tuning parameters in the controller.
- To design and simulate a integral controller to meet the required specifications.
- To implement your controller and evaluate its performance.

! *System requirements are same as Experiment 1.*

MATHEMATICAL MODEL

The result in Experiment 1:

$$\frac{\theta_l(s)}{V_m(s)} = \frac{\eta_g \eta_m K_t K_g}{J_{eq} R_m s^2 + (B_{eq} R_m + \eta_g \eta_m K_m K_t K_g^2)s}$$

The model of the DC motor was derived in Experiment 1 considering the angular position of the load as output. Since $s\theta_l = \omega_l$.

$$\boxed{\frac{\omega_l(s)}{V_m(s)} = \frac{\eta_g \eta_m K_t K_g}{J_{eq} R_m s + (B_{eq} R_m + \eta_g \eta_m K_m K_t K_g^2)}}$$

CONTROLLER

This lab requires you to design a controller to control the angular speed of the load shaft with the desired specifications.

Desired specifications are:

- The system should have zero steady-state error (for a step input).
- The system should have fastest response without over-shoot (critically damped).

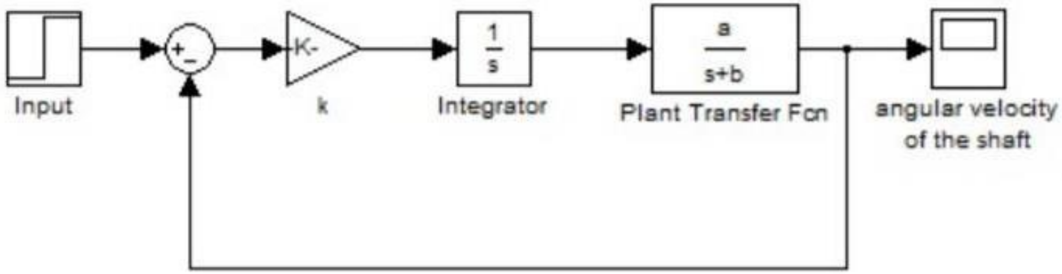

Fig. 5 The block diagram of the closed loop system

CONTROLLER

The purpose of this lab is to design a controller using the 2nd order transfer function representation of the form given:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$ *closed loop transfer function*

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$ *characteristic equation*

! ***Show the steps for finding the coefficients of controller k to meet desired specifications in the experiment report.***

★ ***Hint: You can use root locus approach or pole placement***